

Hypothesis Testing

1. First: What problem does hypothesis testing solve?

Imagine you run an online store.

You currently know:

The average delivery time is **5 days**.

You change your logistics system and after collecting data from 100 orders, you find:

Average delivery time = **4.6 days**

You now have a question:

Did the new system genuinely make deliveries faster, or did we just get lucky with this particular sample?

That's the fundamental problem hypothesis testing solves.

The key idea

We observe a **sample**, but we want to make a conclusion about a **population**.

Population



Take a sample



Collect data



Calculate statistic



Hypothesis test



Conclusion about population

2. The courtroom analogy

This is probably the most important analogy for understanding hypothesis testing.

Imagine a criminal court.

Someone is accused of committing a crime.

The court starts with:

"The defendant is presumed innocent."

The prosecution then presents evidence.

The question isn't:

"Can we prove that the defendant is innocent?"

Instead, the question is:

"Is there enough evidence to reject the assumption of innocence?"

Hypothesis testing works almost exactly the same way.

Courtroom	Hypothesis testing
Defendant	Null hypothesis
"Innocent" assumption	H_0
Evidence	Sample data
Prosecutor	Alternative hypothesis
Standard of evidence	Significance level
Verdict	Reject / fail to reject H_0

So:

H_0 is your starting assumption.

3. The two hypotheses

Every hypothesis test normally starts with two competing hypotheses.

Null hypothesis — H_0

The **null hypothesis** represents the default assumption.

Usually:

There is no effect.

or

There is no difference.

or

Nothing has changed.

Examples:

Medicine

H₀: The new drug has no effect.

Machine learning

H₀: Model A and Model B have the same performance.

Business

H₀: The new website doesn't increase sales.

Agriculture

H₀: Fertilizer A and Fertilizer B produce the same yield.

4. Alternative hypothesis — H_1 / H_a

The alternative hypothesis is what you're trying to find evidence for.

For example:

H₁: The new drug has an effect.

or:

H_1 : The new website increases sales.

or:

H_1 : Fertilizer A produces a different yield.

Think of it this way:

$H_0 \rightarrow$ "Nothing special is happening."

$H_1 \rightarrow$ "Something is happening."

5. An important subtlety

You don't usually say:

"I proved H_0 is true."

Instead, you say:

Reject H_0

or

Fail to reject H_0

This distinction is extremely important.

Why?

Suppose you test whether a coin is fair.

You toss it 10 times and get:

6 heads, 4 tails.

You don't conclude:

"I have proven the coin is fair."

You simply don't have enough evidence to conclude that it's unfair.

That's:

Fail to reject H_0 .

6. The courtroom again

Imagine:

H_0 :

Defendant is innocent.

You examine the evidence.

Strong evidence

You say:

"There is enough evidence to reject the assumption of innocence."

Weak evidence

You say:

"There isn't enough evidence to reject the assumption."

Notice:

Weak evidence $\neq$ proof of innocence.

Likewise:

Fail to reject $H_0 \neq$ prove H_0 is true.

7. What is a test statistic?

Now we get into the mathematics.

Suppose a company claims:

Average battery life = 10 hours.

You test 100 batteries.

Your sample average is:

9.6 hours.

Is 9.6 significantly different from 10?

We need a way to measure:

How far away is our observation from what H_0 predicts?

That's what the **test statistic** does.

A simplified idea is:

$$\text{Test Statistic} = \frac{\text{Observed} - \text{Expected}}{\text{Standard Error}}$$

In plain English:

How many standard errors away from the null hypothesis is our result?

8. The "how weird is this?" analogy

Suppose your friend says:

"I can predict coin flips."

You say:

"Okay, predict 10."

He predicts:

H H H H H H H H H H

And then somehow gets all 10 correct.

You'd probably think:

"Something suspicious is happening."

Why?

Because the result is **very unusual under the assumption that he's just guessing**.

That's the heart of hypothesis testing:

How unusual is our observed data if H_0 were actually true?

9. The null distribution

This is where statistics becomes more interesting.

Suppose:

H_0 : The coin is fair.

We can imagine repeatedly tossing the coin 10 times.

We might get:

5 heads

6 heads

4 heads

7 heads

3 heads

5 heads

...

Most results are around 5.

Very extreme results are rare:

0 heads

1 head

9 heads

10 heads

This collection of possible outcomes is related to the **null distribution**.

It answers:

"What would results normally look like if H_0 were true?"

10. P-value

Now we reach one of the most misunderstood concepts in statistics.

The **p-value** answers approximately:

If H_0 were true, how surprising would our observed result—or something even more extreme—be?

That's it.

Not:

"Probability that H_0 is true."

Not:

"Probability that our result happened by chance."

Those are common but incorrect interpretations.

11. P-value analogy: the suspicious teacher

Imagine a teacher says:

"This class's average score is 60."

You randomly select 50 students and discover:

Average = 61.

That's not particularly surprising.

But suppose you get:

Average = 95.

You'd think:

"Whoa. If the true average really were 60, getting 95 would be extremely unusual."

Therefore:

Small p-value

The result would be unusual under H_0 .

Large p-value

The result isn't particularly unusual under H_0 .

12. The famous 0.05 threshold

A common significance level is:

$$\alpha=0.05$$

This means we're willing to tolerate about a **5% false-positive rate** under the test's assumptions.

Decision rule:

$$p < 0.05$$

→ **Reject H_0**

while:

$$p \geq 0.05$$

→ **Fail to reject H_0**

Example:

$$p = 0.02$$

$$0.02 < 0.05$$

Reject H_0 .

$$p = 0.37$$

$$0.37 > 0.05$$

Fail to reject H_0 .

13. What does "statistically significant" mean?

If:

$$p < 0.05$$

we often say:

"The result is statistically significant at the 5% significance level."

It means:

The observed result would be relatively unusual if H_0 were true.

It **doesn't necessarily mean the effect is important.**

This distinction matters enormously in data science.

14. Statistical significance vs practical significance

Imagine you have 1 million customers.

You introduce a new checkout button.

Sales increase by:

0.01%

Because your sample is enormous, you might get:

$p < 0.001$

Statistically significant?

Yes.

Practically meaningful?

Maybe **not at all**.

This is why you should also examine:

Effect size

15. Effect size

Effect size asks:

How big is the difference?

Suppose:

Model A accuracy:

90.0%

Model B accuracy:

90.1%

With a huge dataset, this could be statistically significant.

But the actual improvement is:

0.1 percentage points.

Compare that with:

90% → 96%

That's a much larger effect.

So always think:

Statistical significance

+

Effect size

+

Practical importance

16. Type I error

Now imagine you conclude:

"The new drug works!"

But actually:

The drug doesn't work.

You've rejected a true H_0 .

That's a:

Type I error

Also called:

False positive

Think:

You accused an innocent person.

17. Type II error

Now imagine:

The drug really works.

But your experiment fails to detect the effect.

You say:

"We don't have enough evidence."

That's:

Type II error

Also called:

False negative

Think:

A guilty person walks free.

18. The confusion matrix analogy

You can connect hypothesis testing to classification.

Reality	Your conclusion	Result
H_0 true	Reject H_0	Type I error
H_0 true	Fail to reject H_0	Correct
H_0 false	Reject H_0	Correct
H_0 false	Fail to reject H_0	Type II error

This is basically the statistical version of:

Reality		
	H_0	H_1
Decision		
H_0	TN	FN
H_1	FP	TP

19. Statistical power

Power is:

$$\text{Power} = 1 - \beta$$

where β is the probability of Type II error.

So power answers:

If a real effect exists, how likely is my test to detect it?

Example:

$$\text{Power} = 0.80$$

means:

If the specified effect really exists, there's about an 80% chance the test will detect it under the stated assumptions.

A common target is:

80% power

20. What determines statistical power?

Power increases when:

1. Sample size increases

More data → easier to detect effects.

2. Effect size increases

A huge difference is easier to detect.

3. Noise decreases

Cleaner data → easier to detect signal.

4. Significance level increases

For example:

0.01 → 0.05

makes it easier to reject H_0 , though it also increases Type I error risk.

21. The signal-to-noise analogy

Think about trying to hear someone in a noisy room.

Small signal + lots of noise

Very difficult.

Large signal + little noise

Easy.

Hypothesis testing is essentially asking:

Is the signal large enough compared with the noise that we're confident it's real?

That's why you'll often see:

$$\frac{\text{effect}}{\text{standard error}}$$

22. Standard error

Standard deviation describes:

How spread out individual observations are.

Standard error describes:

How uncertain our estimate of the population parameter is.

For a sample mean:

$$SE = \frac{s}{\sqrt{n}}$$

where:

- s = sample standard deviation
- n = sample size

Notice:

$n \uparrow \Rightarrow SE \downarrow$

So larger samples give more precise estimates.

23. Why does $\sqrt{n}$ appear?

Imagine you're trying to estimate the average height of Nigerians.

One person gives you a terrible estimate.

100 people gives you a much better estimate.

10,000 people gives you an even better estimate.

But precision doesn't increase linearly.

The standard error shrinks approximately according to:

$$\frac{1}{\sqrt{n}}$$

So:

n Relative SE	
1	1
4	0.5
100	0.1
10,000	0.01

To reduce SE by 10×, you generally need 100× the sample size.

24. Confidence intervals

Hypothesis testing is closely related to **confidence intervals**.

Suppose you estimate:

$$\bar{x}=50$$

and your 95% confidence interval is:

[47,53]

This tells you about the uncertainty around your estimate.

A useful relationship:

For many standard two-sided tests:

If the null value is outside the 95% confidence interval, the corresponding test is significant at $\alpha = 0.05$.

Example:

H₀:

$\mu=50$

CI:

[52,58]

50 isn't inside the interval.

Therefore:

Reject H_0 at approximately the 5% level.

25. One-tailed vs two-tailed tests

This is another major concept.

Suppose you're testing a new fertilizer.

You ask:

Does it change crop yield?

You don't care whether it increases or decreases.

That's:

Two-tailed test

$$H_0 : \mu = \mu_0$$

$$H_1 : \mu \neq \mu_0$$

But suppose you specifically ask:

Does fertilizer increase crop yield?

Then:

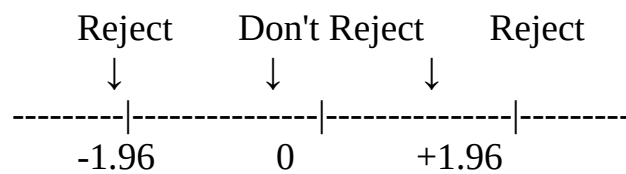
$$H_1: \mu > \mu_0$$

That's a:

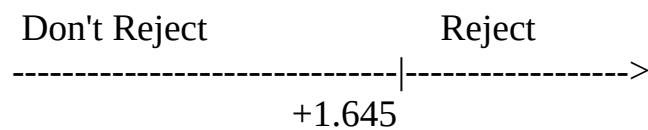
One-tailed test

26. Visualizing it

Two-tailed:



One-tailed:



For $\alpha = 0.05$:

Two-tailed

5% is split:

$$2.5\% + 2.5\%$$

One-tailed

All 5% goes into one tail.

27. When should you use one-tailed tests?

Use one when the alternative hypothesis is genuinely directional.

For example:

$$H_1: \mu > 50$$

Not:

"I'll use one-tailed because it gives me a smaller p-value."

You should decide the direction **before looking at the results**.

28. Z-test

The Z-test is one of the classic hypothesis tests.

For a population mean with known population standard deviation:

$$z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

Where:

- $\bar{x}$ = sample mean
 - μ_0 = hypothesized population mean
 - σ = population standard deviation
 - n = sample size
-

29. Example Z-test

A company claims:

Average delivery time = 10 days.

You collect:

$$n=100$$

Sample mean:

$$\bar{x}=9.5$$

Population standard deviation:

$$\sigma=2$$

H_0 :

$$\mu=10$$

Calculate:

$$z = \frac{9.5 - 10}{2/\sqrt{100}}$$

$$z = \frac{-0.5}{0.2}$$

$$z = -2.5$$

me.

A z-score of -2.5 is fairly extreme.

For a two-sided test, the p-value is around:

$$p \approx 0.012$$

Since:

$$0.012 < 0.05$$

we reject H_0 .

Conclusion:

There is statistically significant evidence that the average delivery time differs from 10 days.

30. T-test

In real-world data science, the **t-test** is often more useful.

Why?

Because we usually don't know the true population standard deviation.

The one-sample t-statistic is:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

The key difference:

Z-test $\rightarrow$ population σ known

T-test $\rightarrow$ population σ unknown; use sample s

31. Why does the t-distribution exist?

This is a beautiful concept.

Suppose you don't know the true population variability.

You're estimating it from the same sample.

That introduces additional uncertainty.

So instead of the standard normal distribution, we use the **t-distribution**.

It has heavier tails.

As sample size increases:

t-distribution $\rightarrow$ normal distribution

32. Degrees of freedom

For a one-sample t-test:

$$df = n - 1$$

Why?

Imagine you have 5 numbers and know their mean.

If you choose the first four freely, the fifth number is constrained because the mean must remain fixed.

So you effectively have:

$$5-1=4$$

degrees of freedom.

33. Independent samples t-test

Suppose you have two groups.

Group A

New teaching method.

Group B

Old teaching method.

You want to know:

Is their average exam score different?

You can use an:

Independent two-sample t-test

Example:

New method:

72, 75, 80, 77, 74...

Old method:

65, 68, 70, 64, 69...

34. Paired t-test

Now suppose you measure the **same students twice**.

Before training:

60

65

70

...

After training:

70

72

78

...

These observations are connected.

Therefore use:

Paired t-test

The test essentially works on the differences:

$$d_i = \text{After}_i - \text{Before}_i = \text{After}_i - \text{Before}_i$$

Then tests whether:

$$\mu_d = 0 \quad \mu_d = 0$$

35. A simple analogy

Independent test

Compare:

Class A vs Class B

Different people.

Paired test

Compare:

You before training vs you after training.

Same person.

Remember:

Same subjects → think paired.

36. Chi-square test

Now suppose your variables are categorical.

Example:

You want to know whether:

Gender and preference for an AI product are associated.

You might have:

	Like	Don't Like
Male	60	40
Female	40	60

The **chi-square test of independence** asks:

Is there an association between these categorical variables?

37. Chi-square intuition

Imagine you're sorting colored balls into boxes.

If there's no relationship, the distribution should look roughly like what we'd expect by chance.

Chi-square compares:

Observed counts

against:

Expected counts

The formula is:

$$\chi^2 = \sum (O - E)^2 / E$$

where:

- O = observed
- E = expected

Big differences between O and E produce a large χ^2 statistic.

38. ANOVA

Suppose you have:

Model A

Model B

Model C

Model D

and want to know whether their mean performances differ.

Instead of doing many t-tests, you can use:

ANOVA

ANOVA stands for:

Analysis of Variance.

It essentially asks:

Is the variation between groups large relative to the variation within groups?

39. ANOVA analogy

Imagine four classrooms.

Each classroom has students with different exam scores.

You observe:

Class A → 70, 72, 71

Class B → 71, 73, 72

Class C → 90, 92, 91

Class D → 69, 70, 71

Class C looks very different.

ANOVA asks:

Is the difference between classroom averages too large to reasonably attribute to random variation?

40. The F-statistic

ANOVA uses:

$$F = \frac{\text{Between-group variability}}{\text{Within-group variability}}$$

This is an incredibly useful intuition.

Large F

Groups differ substantially compared with their internal noise.

Small F

Groups look relatively similar.

41. Important ANOVA trap

Suppose ANOVA gives:

$$p < 0.05$$

You can conclude:

At least one group differs.

But you **cannot immediately say which groups differ**.

You need a post-hoc test, such as:

Tukey's HSD.

42. Correlation hypothesis testing

Suppose you calculate:

$$r = 0.72$$

between:

Study hours and exam score.

You might test:

$$H_0: \rho = 0$$

against:

$$H_1: \rho \neq 0$$

You're asking:

Is the observed correlation statistically distinguishable from zero?

But remember:

Correlation $\neq$ causation.

43. Regression hypothesis testing

This is extremely important for data science.

Suppose:

$$\text{Salary} = \beta_0 + \beta_1 \text{Experience} + \epsilon \quad \text{Salary} = \backslash\text{beta_0} + \backslash\text{beta_1 Experience} + \backslash\text{epsilon}$$

You want to know whether experience predicts salary.

You test:

$$H_0: \beta_1 = 0 \quad H_0: \backslash\text{beta_1} = 0$$

against:

$$H_1: \beta_1 \neq 0 \quad H_1: \backslash\text{beta_1} \neq 0$$

If the coefficient's p-value is small, you have evidence that the coefficient differs from zero under the model assumptions.

44. Testing machine-learning models

This is particularly relevant to you as an ML engineer.

Suppose:

Model A

Accuracy = 91%

Model B

Accuracy = 93%

You shouldn't automatically say:

"Model B is better."

You need to ask:

Is the difference statistically meaningful?

Depending on the evaluation setup, you could use methods such as:

- paired tests
- bootstrap confidence intervals
- permutation tests
- McNemar's test for paired classification outcomes
- tests on cross-validation results, with care
- DeLong's test for comparing correlated ROC AUCs in suitable settings

The correct test depends on **what data you're comparing**.

45. McNemar's test

This is particularly useful for classification.

Suppose two classifiers make predictions on the **same test examples**.

You care about where they disagree.

Example:

	Model B correct	Model B wrong
Model A correct	800	100
Model A wrong	150	50

McNemar's test focuses on the discordant pairs:

A correct / B wrong

A wrong / B correct

It asks whether those disagreement rates are significantly different.

That's often more appropriate than blindly doing a t-test on accuracy.

46. Non-parametric tests

What if your data doesn't satisfy assumptions needed for classical tests?

Then you may use non-parametric alternatives.

Common ones:

Parametric	Non-parametric alternative
One-sample t-test	Wilcoxon signed-rank
Independent t-test	Mann–Whitney U
Paired t-test	Wilcoxon signed-rank
One-way ANOVA	Kruskal–Wallis
Repeated-measures ANOVA	Friedman test

These often work with fewer distributional assumptions.

But don't think:

"Non-parametric = no assumptions."

They still have assumptions.

47. Mann–Whitney U

Suppose you compare two independent groups.

Your data is heavily skewed:

Group A:

2, 3, 3, 4, 1000

Group B:

5, 6, 7, 8, 9

A Mann–Whitney U test may be more appropriate than a standard t-test depending on the question and assumptions.

It works with **ranks** rather than relying on the same mean-based assumptions as the t-test.

48. Wilcoxon signed-rank test

Suppose you have paired observations:

Before → After

but the differences don't satisfy assumptions for a paired t-test.

You can consider:

Wilcoxon signed-rank test.

It analyzes the ranks of the paired differences.

49. Kruskal–Wallis

Think:

"ANOVA, but rank-based."

You have three or more independent groups and want to compare their distributions/location.

Example:

Treatment A

Treatment B

Treatment C

Treatment D

If ANOVA's assumptions aren't appropriate, Kruskal–Wallis may be considered.

50. Assumptions matter

A hypothesis test isn't a magic button.

You need to understand its assumptions.

For example, depending on the test:

t-test assumptions can include

- independent observations
- appropriate measurement scale
- approximately normal sampling distribution / normality of relevant data in small samples
- appropriate handling of outliers
- independence

For two-sample tests, you also need to consider variance assumptions—or use **Welch's t-test**, which is generally safer when variances differ.

51. Normality: an important clarification

People often say:

"Your data must be perfectly normal for a t-test."

That's too simplistic.

The t-test concerns the **sampling distribution of the mean** and its assumptions.

With sufficiently large samples, the Central Limit Theorem often makes the sample mean approximately normal even when individual observations aren't perfectly normal.

However:

Severe skewness, outliers, dependence, or small sample sizes can still cause problems.

So don't blindly test "normality" and stop there.

52. Central Limit Theorem

This is one of the foundations of statistical inference.

Suppose the underlying population isn't normal.

You repeatedly take samples and calculate their means.

As sample size increases, the distribution of those sample means tends toward a normal distribution under broad conditions.

Conceptually:

Messy population



Take many samples



Calculate each sample mean



Distribution of means



Approximately normal

That's a major reason statistical inference works.

53. The entire hypothesis-testing pipeline

Here's the process you should memorize.

Step 1 — Define the question

Example:

Does the new model improve accuracy?

Step 2 — Define H_0

H_0 :no improvement $H_0:\text{no improvement}$

Step 3 — Define H_1

H_1 :improvement $H_1:\text{improvement}$

Step 4 — Choose α

Usually:

$\alpha=0.05$ $\alpha=0.05$

Step 5 — Choose the appropriate test

Could be:

- t-test
- z-test
- chi-square
- ANOVA
- Mann–Whitney
- Wilcoxon
- permutation test
- etc.

Step 6 — Calculate test statistic

Step 7 — Calculate p-value

Step 8 — Make decision

$p < \alpha$

↓

Reject H_0

$p \geq \alpha$

↓

Fail to reject H_0

Step 9 — Interpret in context

Don't just say:

$p = 0.023$.

Say:

"There is statistically significant evidence that the new model improves performance."

54. The p-value doesn't tell you the probability H_0 is true

This deserves repetition.

Suppose:

$p=0.03$

It does **NOT** mean:

"There's a 3% chance H_0 is true."

It means roughly:

If H_0 were true, results at least this extreme would occur about 3% of the time under the test's assumptions.

That's fundamentally different.

55. Statistical significance is not certainty

Suppose:

$p=0.049$

and another experiment gives:

$p=0.051$

Don't think:

0.049 → discovery!
0.051 → nothing happened!

They're practically almost identical.

The 0.05 threshold is a convention, not a magical boundary between truth and falsehood.

56. Multiple hypothesis testing

Now imagine you test:

100 different hypotheses.

You use:

$$\alpha = 0.05$$

Even if **none of the hypotheses is actually true**, you'd expect approximately:

$$100 \times 0.05 = 5$$

false positives on average under the idealized setup.

This is called the:

Multiple comparisons problem.

57. Bonferroni correction

One simple correction is:

$$\alpha_{\text{adjusted}} = \frac{\alpha}{m}$$

where m is the number of tests.

Suppose:

$$\alpha=0.05 \backslash \alpha=0.05$$

and:

$$m=10 m=10$$

Then:

$$\alpha_{\text{adjusted}}=0.05/10=0.005 \backslash \alpha_{\text{adjusted}} = \frac{0.05}{10} = 0.005$$

Now you'd require:

$$p < 0.005 p < 0.005$$

for significance under Bonferroni.

It's simple and conservative.

58. False Discovery Rate

In data science, you may encounter:

FDR — False Discovery Rate

A famous procedure is:

Benjamini–Hochberg

This is useful when conducting many statistical tests and trying to control the expected proportion of false discoveries among the rejected hypotheses.

59. Bayesian hypothesis testing

Everything we've discussed so far is primarily **frequentist statistics**.

Bayesian statistics takes a different perspective.

Frequentist:

"How surprising is this data assuming H_0 ?"

Bayesian:

"Given the data, how should I update my belief about the hypothesis?"

Bayes' theorem:

$$P(H|D) = \frac{P(D|H)P(H)}{P(D)}$$

Where:

- $P(H)$ = prior belief
 - $P(D|H)$ = likelihood
 - $P(H|D)$ = posterior belief
-

60. Bayesian analogy

Imagine you hear:

"A disease test came back positive."

Before testing, you know the disease is rare.

That's your:

Prior

Then you see the positive test.

That's your:

Evidence

You update your belief.

That's your:

Posterior

This is conceptually different from simply calculating a p-value.

61. Permutation tests

Permutation testing is incredibly useful and intuitive.

Suppose you have:

Group A

10, 12, 14, 15, 13

Group B

20, 22, 18, 21, 23

You observe a difference in means.

The permutation idea asks:

What if group labels were completely arbitrary?

You randomly shuffle the labels thousands of times.

For every shuffle, calculate the difference.

You build a distribution of differences that would occur if there were **no real group effect**.

Then ask:

How often do we get a difference as extreme as the observed one?

That's your empirical p-value.

62. Bootstrap

Bootstrap is related but answers a somewhat different question.

You have:

Data:

10, 12, 15, 17, 20

You repeatedly sample **with replacement** from these observations.

For example:

10, 10, 15, 20, 20

Then:

12, 15, 15, 17, 20

and so on.

Calculate the statistic each time.

This gives you an empirical sampling distribution.

You can use it to estimate:

- standard errors
- confidence intervals
- uncertainty
- sometimes hypothesis tests

63. Bootstrap vs permutation

A useful mental distinction:

Bootstrap

"How uncertain is my estimate?"

Permutation

"How compatible is this difference with the idea that labels don't matter?"

They can overlap in applications, but conceptually they're different tools.

64. A complete real-world example

Let's say you develop two ML models for mammogram classification.

Model A

F1:

0.720.72

Model B

F1:

0.760.76

You want to know:

Is Model B genuinely better?

You shouldn't simply compare:

$0.76 > 0.72$

Instead, formulate a statistical question.

Suppose you evaluate both models on the same cases repeatedly using a suitable resampling procedure.

H₀

H₀: No difference in performance

H₁

H₁: Performance differs

Suppose your analysis produces:

$p = 0.018$

At:

$\alpha=0.05$

we reject H_0 .

You might report:

"The difference in model performance was statistically significant ($p = 0.018$)."

But don't stop there.

You should also report:

effect size + confidence interval + evaluation methodology.

65. Why accuracy alone can be dangerous

Suppose you have:

Normal 90%

Benign 5%

Malignant 5%

A model predicts:

Normal

for every image.

Accuracy:

90%

Looks impressive.

But the model detects:

0% malignant cases.

That's terrible clinically.

This is why in classification we often examine:

- precision
- recall
- F1
- sensitivity
- specificity
- ROC-AUC
- PR-AUC
- confusion matrix

And statistical testing should respect the **paired nature and structure of the evaluation data**.

66. Sensitivity and specificity

For medical classification:

Sensitivity

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

It answers:

Of the people who actually have the disease, how many did we detect?

Specificity

$$\text{Specificity} = \frac{TN}{TN + FP}$$

It answers:

Of the people who don't have the disease, how many did we correctly identify?

67. Type I and II errors in medical AI

This makes the earlier concept very concrete.

False positive

Model says:

"Cancer."

Patient doesn't have cancer.

That's related to:

Type I error in a simplified hypothesis-testing analogy.

False negative

Model says:

"No cancer."

Patient actually has cancer.

That's related to:

Type II error in the same simplified analogy.

In medicine, the costs of these errors can be very different.

That's why α isn't just a mathematical choice.

68. Hypothesis testing vs confidence intervals

Think of them as two sides of the same coin.

Hypothesis test

Asks:

"Is this value compatible with the data?"

Confidence interval

Asks:

"What range of values is reasonably compatible with the data?"

For example:

Difference=4.2

95% CI:

[1.1, 7.3]

This is much more informative than simply saying:

$p = 0.01$.

69. The hierarchy you should remember

When doing statistical analysis, think:

QUESTION



HYPOTHESES



DATA



ASSUMPTIONS



TEST



TEST STATISTIC



P-VALUE



CONFIDENCE INTERVAL



EFFECT SIZE

↓
PRACTICAL INTERPRETATION

That is much better than:

Run test → look for $p < 0.05$

70. The most common hypothesis tests

Here's a cheat sheet.

Situation	Typical test
One mean vs known value	One-sample t-test
Two independent means	Independent t-test / Welch's t-test
Same subjects before/after	Paired t-test
3+ independent means	ANOVA
3+ repeated measurements	Repeated-measures ANOVA
Two categorical variables	Chi-square independence
One categorical distribution	Chi-square goodness-of-fit
Correlation	Pearson correlation test / alternatives
Two independent non-normal groups	Mann–Whitney U
Two paired non-normal groups	Wilcoxon signed-rank
3+ independent non-normal groups	Kruskal–Wallis
3+ paired non-normal groups	Friedman
Paired binary predictions	McNemar
Flexible null testing	Permutation test

71. A decision tree

When you face a new problem:

Question 1

What type of variable am I dealing with?

Continuous?

Categorical?
Binary?
Ordinal?

Question 2

How many groups?

1
2
3+

Question 3

Are groups independent or paired?

Independent
or
Same subjects / matched

Question 4

What exactly am I testing?

Mean?
Proportion?
Association?
Distribution?
Model performance?
Correlation?

Then choose the test.

72. Common mistakes

Mistake 1

" $p = 0.03$ means there's a 3% probability H_0 is true."

✗ Wrong.

Mistake 2

" $p > 0.05$ proves H_0 ."

✗ Wrong.

Correct:

Fail to reject H_0 .

Mistake 3

"Statistically significant means important."

✗ Not necessarily.

Mistake 4

"I'll choose one-tailed after seeing the data."

✗ Bad practice.

Mistake 5

"I ran 50 tests and found $p < 0.05$, so I found something."

⚠ Multiple testing problem.

Mistake 6

"I'll use a t-test for everything."

✗ Test choice depends on the question and data structure.

Mistake 7

"Correlation proves causation."

✗ Absolutely not.

73. The deepest intuition

If you remember only one equation, remember the conceptual structure:

$$\text{Evidence} = \frac{\text{Observed Effect}}{\text{Uncertainty}}$$

A large effect isn't necessarily convincing if uncertainty is huge.

A small effect can become statistically significant if uncertainty is tiny.

That's why:

$$\text{Test statistic} \approx \frac{\text{signal}}{\text{noise}}$$

74. One giant analogy tying everything together

Imagine you're a detective.

Someone tells you:

"There is no thief in this building."

That's:

H0

You investigate.

You find a fingerprint.

That's:

Evidence.

You ask:

"How surprising would this fingerprint be if there really were no thief?"

That's essentially the intuition behind the:

p-value.

If the evidence is extremely unusual under H_0 :

Reject H_0 .

If the evidence isn't strong enough:

Fail to reject H_0 .

But then you ask:

"How large is the actual effect?"

That's:

Effect size.

Then:

"How uncertain is my estimate?"

That's:

Confidence interval / standard error.

Then:

"Could my test have missed a real effect?"

That's:

Statistical power / Type II error.

And:

"Could I have falsely detected an effect?"

That's:

Type I error.

And finally:

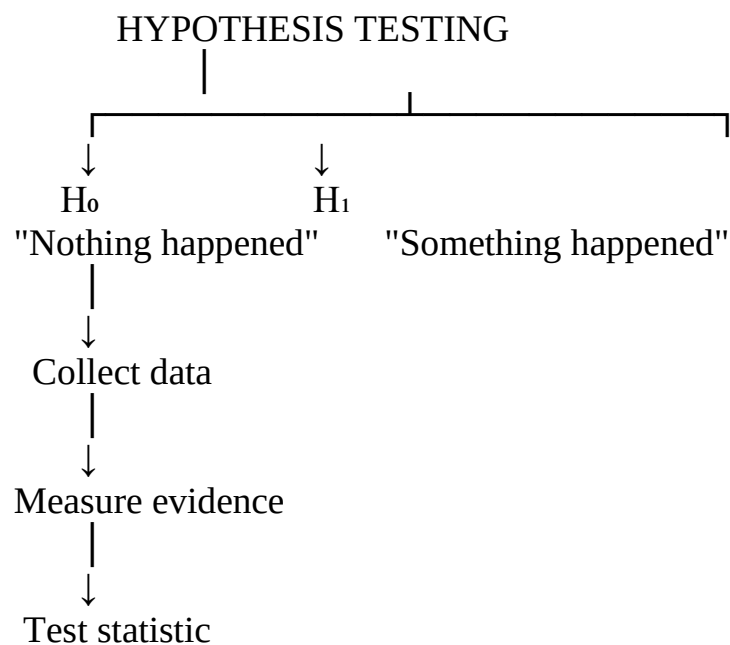
"Did I investigate 100 suspects?"

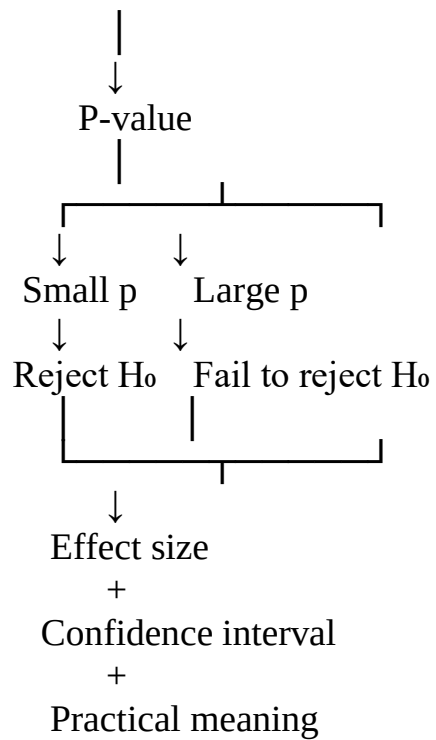
That's:

Multiple testing.

75. Your mental model

I want you to internalize this:





And the **four words** I most want you to remember are:

Null → Evidence → P-value → Decision